Resonate App - GSOC Proposal

Google Summer Of Code 2024 Proposal

Top Project Of Choice:

My top priority would be Resonate App. I would like to implement chat functionality in rooms and a friends system where you can chat with and call up your friends, along with UI changes such as perfecting the currently implemented app colour feature, and reconfiguring the way app fetches data and shows UI for it in order to improve UX. Apart from this completing the current PR i am working and deploying app to play store

Are you willing and able to work on other projects instead?

I am willing to work on other projects from AOSSIE.

Please describe your preferred coding languages and experience.

I have explored and worked in couple of domains. Flutter App Development and Server Side Development (backend development) are the most prominent of them. I have some experience working with docker as well.

- Flutter/Dart

  I have been into flutter developer since around 2 years now, here are some of my near to complete projects (4th still in progress)

  - ResumeUp - A dynamic portfolio builder

    https://github.com/TiDB-Hacks/ResumUp

  - Sustainify - A hub for waste management on a locality level

    https://github.com/Flyer-Smart-India-Hackathon-2023/Sustainify

  - Isvara - An AI-based software solution for constant eye strain from prolonged screen time

    https://github.com/Aarush-Acharya/Isvara-Frontend

  - TaskWatch - `In progress`, a productivity analyser fetching user data from different platforms like Github and other websites

    https://github.com/Aarush-Acharya/taskwatch.git

  - Water Works - `In progress`, an app for spreading awareness and take communal action against water pollution moving forward to saving local water bodies

    https://github.com/Aarush-Acharya/waterworks.git

- Python

Started programming in python in 2022, built several API's using Fast API, and Flask

- Resume Up Backend - The Flask Backend of the main Resume Up website which is responsible for initiating the template deployment once the user chooses to deploy its portfolio. It uses TiDB serverless SQL DB

  https://github.com/TiDB-Hacks/ResumeUp_Server

- Isvara Server - Server Side for Isvara implemented in Flask, responsible for returning distance of a person from the camera in the picture sent via a POST request

  https://github.com/Isvaraa/Isvara-Server

- Sarover Server - Handles. prompt answering via Bard API

  https://github.com/Aarush-Acharya/Sarover-Server

- **TypeScript**
  - Resume-Up Template Vercel Edge Functions - The API's written as Vercel Edge Functions of the template which was created to be put up to users Vercel once the user designs and configures it in the ResumeUp create portfolio section. The functions were responsible for fetching the users Github and Vercel info (activity and deployments) into the template.

- JavaScript
  - Resonate Appwrite Cloud Functions (Node.js runtime) - 4 Cloud Functions (Send OTP, Verify Email, Verify OTP, DiscussionIsTimeChecker) for Resonate project of AOSSIE organisation

Resonate-Backend/functions/verify-otp at main · AOSSIE-Org/Resonate-Backend

Appwrite Cloud Functions for Resonate Project. Contribute to AOSSIE-Org/Resonate-Backend development by creating an account on GitHub.

https://github.com/AOSSIE-Org/Resonate-Backend/tree/main/functions/verify-otp

AOSSIE-Org/**Resonate-Backend**

Appwrite Cloud Functions for Resonate Project.

8 Contributors · 0 Issues · 16 Stars · 56 Forks

Resonate-Backend/functions/verify-email at main · AOSSIE-Org/Resonate-Backend

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Personal Information Details

Please describe any Windows, Unix or Mac OS X development experience relevant to your chosen project. If your project does not require OS-specific expertise, feel free to leave it empty.

I have a Mac M1 Laptop, hence i can build and test, the application on both IOS and Android operating systems, making sure it works perfectly on both

Please describe any previous AOSSIE Project or AOSSIE related development experience, including details of any patches, code or ideas you may have previously submitted.

PR	Status
https://github.com/AOSSIE-Org/Resonate/pull/208	Merged
https://github.com/AOSSIE-Org/Resonate/pull/115	Merged

https://github.com/AOSSIE-Org/Resonate/pull/105	Merged
https://github.com/AOSSIE-Org/Resonate/pull/85	Merged
https://github.com/AOSSIE-Org/Resonate/pull/69	Merged
https://github.com/AOSSIE-Org/Resonate-Backend/pull/29	Merged
https://github.com/AOSSIE-Org/Resonate-Backend/pull/42	Merged

I also worked as a Software Developer Intern at AOSSIE, from Sept 14 2023 - Oct 23 2023, as the work of my tenure I implemented Discussions part of Resonate include the Backend Logic and the UI Screens for it.

What school do you attend and what is your specialty/major at the school?

I am currently attending Bennett University (Greater Noida, UP, India), pursuing my B.Tech in CSE from there. My specialisation is in Data Science.

How many years have you attended there?

Almost 3 years by now. My course is for 4 years.

What city/country will you be spending this summer in?

Noida, UP mostly.

How much time do you expect to give per day to this project?

I would be able to devote 5-6 hours per day during my summer vacations being 25-30 hours a week and 25-30 hours per week.

I'll be having 6th semester End Semester Examinations around April end so they would end prior to the start of GSOC tenure, and then my vacations start up, and end around July end.

Though, My university happens to prioritise programs like GSoC and is open to supporting me accordingly if my proposal is selected.

Have you participated in any previous Summer of Code projects? If so please describe your projects and experience, including what you liked or didn't like about the experience

I haven't. This is my first time.

Have you applied for (or intend to apply for) any other Google Summer of Code 2023 projects? If so, which ones?

I haven't and I do not intend to do the same. I am, however open to working at other AOSSIE Projects.

If you have a URL for your resume/CV, please list it here.

here is my resume

Project details

I would primarily like to focus on overall development of the Resonate App, adding up new functionalities, with some adjustments to currently implemented UI features and bug fixes, and its deployment to play store.

I plan on accomplishing the following tasks.e

Tasks

1. Integrate Chat/Comments functionality to Rooms/Discussions
    - Add Tagging and notification on tagged
    - Text Widget
    - Chat Screen
2. Reconfigure Pair Chat Experience
    - New Pair Chat Info Screen
        - List of users in the lobby (along with count on the top)
    - Business logic for choice whom to pair with from lobby
    - Ratings of users collected after pair chat Experience
3. Implementing a Concept of Friends

- List Friends, Send/Receive Friend Requests
- Chat with Friends
- Call Up Friends (including ringing on receivers)

4. Stories/Podcast
    - Screen to create podcasts
    - Indivisual screen of podcasts and play them
    - A screen to search podcast having categories and all like moods types (Bussiness, self thoughts, random stories)
    - Adding them to favourites and all
    - A feature to view those favourites from users personal account info
5. Enhance App Colour Functionality and Bug Fixes
6. Push to PlayStore and App Store

Expanding on the tasks below

1. Creating a Script to give Dummy Backend Env

The goal of this task is to easy the onboarding process of new contributors and provide them with a script created replica of AOSSIE Resonate's main Appwrite backend instance enabling new contributors to explore all the functionalities of the App.

This would keep the credentials of the main instance private and safe and to be only provided to contributors who have committedly worked on a functionality in the dummy replica instance self hosted by them on their own systems and now need to be pushed into the main system of the Application

2 . Integrate Chat/Comments functionality to Rooms/Discussions

The Current Rooms lack the essence of textual conversation inside the room in the form of comments and reactions

In the Appwrite database collection `rooms` we can add an attribute chat in order to store a list of JSON (Map in Dart) of the following structure

As

```
{
 "author_id": <"User uid">,
 "author_name": <"User Display name">,
 "author_pic_Url": <"User pic Url">,
 "message": <"The Message">
 "reply_to": {
          "author_id": <"User Id being replied to">,
          "author_name": <"User Display name being replied to">,
          "message": <"the message being replied to">
          }
}
```

Use cases of each parameter

`author_id` : To fetch any data of the author

`author_name` : Display the name of author on top of the Text Widget

`author_pic_url` : Display profile pic of the author beside the Text Widget

`message` : Display the actual message

`reply_to` : To render a different Text Widget UI if it is a reply to a prev message along with the data of the prev message being replied to

This functionality will require the creation of a Text Widget in order to **componentise** the application's code base and be in a cleaner state, the Text Widget will take two different forms based on if the current message is a reply to a previous message or not.

This would further include functionalities like tagging other users and so on.

Similar to the Rooms chat feature, discussions should also get a comments/chat functionality this would aid the subscribers to communicate with the hosts and inquire on if the discussion does not start on time and is running late

This functionality could reuse the final Chat Screen widget created while the integration of the Chat Functionality to the Rooms

3. Improve Pair Chat Experience

Currently the Pair Chat functionality lets us just started pair with the other person, This needs to be improved, as once we are into pairing we are not aware of when will we be connected or so.

I propose that a new screen needs to be built and the way to access pair chat need to change
I plan to have pair chat as one of the option in the friends screen that i am about to pitch below,

When selected that Random Pair Chat Option, we can load up a screen, which shows a list of people waiting the lobby to be paired with, this thing needs to be realtime and actively listen to changes, then the user has the option to choose from the list of people appearing available for pair chat or he can just press the shuffle button at the top of the list which would randomly link to any of the other users available

This way the user wanting to pair will have more information on weather or not will he be paired and can even choose whom to pair with or go by random.

We can also include a Ratings System collecting ratings after the end of a pair chat for that other user in the pair and then reflecting those rating in user's profile and the list that we show in the New Pair Chat Screen

This can go up to form a community where you users can search up usernames and with a search feature and follow them for more notifications on if they go up for pair chat.

4. Implementing a Concept of Friends

Resonate a social voice platform as it goes as a voice platform should have the functionality to call specific known people as well, a functionality like pair chat is already implemented which pairs users up with random users to speak with, why not have known friends and talk to them on it.

A users profile would have two options "Follow" the one responsible for notifications on their activities and "Friend" this one is responsible for this feature

Making resonate a friends hangout spot as well, having chat functionalities with users added as friends and the ability to call them up with phone icon in the top of the chat screen, if there is a need to chat up with more then one friend then a room could be created and the link could be shared, the room and the link sharing part is implemented, i propose a one to one chat and call functionality with the users added a friends

This would require the following

- A screen which has three options in the top being
  - Friends

    Showing the list of your current friends and along with opening up the chat screen if tapped on a friends tile
  - Requests

    Being divided into two sections the Top and Bottom, Top shows all the requests received for being friends with the sender in a scroll view fashion, while the Bottom displays the requests

sent by the user to other users for being its friend again in a scroll view fashion

- Rondom Pair Chat

  This is for the Pair Chat feature

- A new collection Requests needs to be formed with the following attributes

```
{
 "sender_id": "<Sender's user id>",
 "sender_username": "<Sender's Username>",
 "sender_display_name": "<Sender's Display Name>"
 "sender_photo_url": "<Sender's Display picture Url >",
 "reciever_id": "<Reciever's user id>",
}
```

  This should be enough to achieve the functionalities associated with the Requests Screen

  For the Friends Screen we can add an attribute to the `users` collection “friends” this would again be a list of JSON i.e Map in dart which would have the following structure

```
{
 "friend_uid": "<Friend's user id>",
 "friend_display_name": "<Friend's Display Name>",
 "friend_photo_url": "<Friend's Display picture Url>",
 "friend_username": "<Friend's Username>""
}
```

  With this we could fetch and display the list of friends along with the display picture and display name of theirs.

- The calling functionality would be handled using `Livekit` for Audio streaming and the ringing part on the receiver’s side needs to be looked into

5. Stories/Podcasts

Resonate as a social voice platform should have a feature that lets user upload their voice to the community in the form of story, series of story etc. As we already touched upon forming a community in Pair Chat Community, this would be an add on to that.

Now as we know a Users account section will have Friend, Follow buttons along with count of friends and followers that person has, once you follow a user then you will also receive notifications if they post a new Voice note in a single stand alone form of a series of voice notes. Each Voice Note/ Podcast will have a cover image title and all.

This feature for UI needs

- Screen to create podcasts
  This will include wether if the post is a list of podcasts or a stand alone, include its title along with the category it belongs and all
- Indivisual screen of podcasts and play them
  This will show the Cover image as a large image below it will be ttitle then the voice notes
- A screen to search podcast having categories and all like moods types (Bussiness, self thoughts, random stories)
  This will have some podcasts/stories there in the form of cards and in the place where it have categories on clicking the categories it will open up a list of podcasts belonging to that list from different users the podcast can be stand alone or series
- Adding feature to add podcasts to favourites and all
- A feature to view those favourites from users personal account info and remove favourites when on the favourites screen

6. Enhance App Colour Functionality and Bug Fixes

Currently in the App its visible that changing the App Colour Functionality works well but still the App has some elements that remain with the default app colour i.e yellow, thus the functionality is not working completely perfectly i would like to make it butter smooth and entirely complete and without any non colour changing elements

Apart from that fixing any bugs around that time and currently reported UI bugs with the goal of bringing the app to the production level final stage, i.e being perfectly ready for deployment with everything being patched and taken care of will be done

7. Push to PlayStore and App Store

The Resonate App will be deployed to PlayStore as v1 (version 1), after studying the deployment process making it accessible to any having an android phone and the app sending and recieving data from our main backend instance

I would propose deploying the App to App Store as well for users having an IOS device, i am familiar with publishing an to Test Flight and working with the apple developer portal. I would make sure the app works

perfectly fine and reaches its production level for IOS devices as well, pushing it to App Store as a v1 (version 1)

The deployment to PlayStore and AppStore of the production level v1 of the App would mark the end of my GSOC tenure as a mentee.

Timeline

Period	Description	Milestone
Bonding Period & Wrap Up Task 1 (May 4 - May 28)	I start off and sprint through the development of script for dummy backend env and get it wrapped up and done (Task one should be done), Start implementation and integration of chat into rooms and get it wrapped up including creation of Text Widget	Get Task 1 and Task 2 Wrapped Up
Week 1 & 2 (May 29 - June 10)	Start Implementing concepts of friends and be done with it, including friend requests and showing friends in friends section	Start off concept of friends
Week 3 (June 12 - June 17)	Look into and plan for the Friends chat and implement it Start looking into the calling of friends and Livekit Audio streaming	Get Done with chat functionality of friends
Week 4 (June 20 - June 24)	Implement LiveKit Audio streaming and the calling after discussion on the methodology to implement and finish it up	Get done with Audio part of Friends Finish Concept of Friends (Task 4 Done)
Week 5 & 6 (June 27 - July 8)	Start with the improving the pair chat experience and get done with it	Start and wrap up Task 3
July 11th - July 25th	Start with the podcasts and stories and wrap it up including community formation	Start and wrap up Task 5
July 25th - July 29th	Working on Enhancing App Colour functionality and smoothening things up making sure the app reaches is production ready and everything is patched and wrapped up	Fix App Colour, Bug Fixes, Enhance App Level
Post first evaluation August 8th - August 12th	Working on any reviews and gaps and getting things perfectly done Pushing the App to PlayStore and AppStore	Work on Feedback Deploy

Deliverables

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Why are you well suited to perform this project and why are you interested in it?

- I have a good understanding of the code-base being a past contributor.
- Being interested in Audio Streaming in general, I understand the vision of this project and where the mentors want to take it.
- My experience in flutter and development principles would help me create a beautiful and powerful social voice platform, plus working on this project would also help brush up my conduct for scalable and well written code bases along with tapping deeper into Livekit and exploring to get the giving a ring to receiver would help me expand my horizons in app dev.